Zadanie 5 [3 pkt] – Typ 1.1

Oblicz $\left[\left(6-2\sqrt{5}\right)^{\frac{1}{2}}-\left(6+20^{\frac{1}{2}}\right)^{\frac{1}{2}}\right]^2$

Zauważmy, że mamy do czynienia ze wzorem skróconego mnożenia

$$(a-b)^2=a^2-2ab+b^2$$

$$\left[\left(6-2\sqrt{5}\right)^{\frac{1}{2}}-\left(6+20^{\frac{1}{2}}\right)^{\frac{1}{2}}\right]^2$$

$$a^{\frac{m}{n}}=\sqrt[n]{a^m}$$

$$\sqrt{20}=2\sqrt{5}$$

$$\left[\left(6-2\sqrt{5}\right)^{\frac{1}{2}}-\left(6+20^{\frac{1}{2}}\right)^{\frac{1}{2}}\right]^2=\left[\left(6-2\sqrt{5}\right)^{\frac{1}{2}}-\left(6+2\sqrt{5}\right)^{\frac{1}{2}}\right]^2=$$

$$=\left(\left(6-2\sqrt{5}\right)^{\frac{1}{2}}\right)^2-2\cdot\underbrace{\left(6-2\sqrt{5}\right)^{\frac{1}{2}}\cdot\left(6+2\sqrt{5}\right)^{\frac{1}{2}}}_{a^2-b^2=(a-b)(a+b)}+\left(\left(6+2\sqrt{5}\right)^{\frac{1}{2}}\right)^2=$$

$$=6-\cancel{2\sqrt{5}}-2\left(36-20\right)^{\frac{1}{2}}+6+\cancel{2\sqrt{5}}=12-2\cdot(16)^{\frac{1}{2}}=12-2\cdot\sqrt{16}=$$

$$=12-2\cdot4=12-8=4$$

Odp: 4.